

Week 7 teaching guide

Add one challenge · GROW computing · 40 minutes

I can add one new feature that changes how the game is played.

AQA 71638 Programming with Scratch unit 6

Before the lesson

Open your own W06_Win.sb3 and save a copy. Have the named booklet, a teacher demo and the approved save location ready.

Teacher example: W07_Teacher_Model.sb3. Pupil access file: W07_Recovery_Win.sb3.

Recovery supplies prior work only and does not evidence authorship.

Independent Task: allow 25-32 for making and testing, then 32-35 for an individual demonstration and pupil voice.

Stage and time	Teach and check
Arrival / 0-3	Open your own W06_Win.sb3. Save a copy as W07_Challenge.sb3.
Starter / 3-6	A new background only looks different. A moving obstacle changes the route.
I Do / 6-11	Create a new sprite and name it Hazard. Give it two glide blocks inside forever.
We Do / 11-18	Predict a meeting at x 0, y 0, then test avoidance at x 30, y 0. Test a run while the Hazard moves.
I Do 2 / 18-21	The consequence belongs to Player. Player checks touching Hazard?
We Do 2 · Task / 21-25	Test a run with no contact. Test the narrow gap as the Hazard passes.
Independent Task / 25-35	Build a moving Hazard and a contact rule on Player. Test the changed game twice.
Exit / 35-40	Save as W07_Challenge.sb3. Load it and show the new feature.

Choose the task route

Step by step: Build a moving Hazard and its contact rule, one step at a time.

Main task: Build both scripts. Test contact, avoidance and a fresh run.

Challenge: After the core task, compare 2-second and 4-second glides.

Teaching notes and answers

Use the teacher model for demonstration, then return to pupil construction

Adding a picture is not adding a mechanic. Ask: what must the player now do that they did not do before? If nothing changes, the feature is decoration.

Block and design reference

Find this	Use it like this
First	Save a copy before changing anything. W06_Win.sb3 stays as the working game.
New sprite	Choose a Sprite, then Paint or the library. Rename it in the sprite pane, not in the code.
Movement	glide [2] secs to x y, twice, inside forever. Compare both timings from the same positions; slower is not always easier.
Consequence	The check belongs to the sprite that must react. Player checks touching [Hazard]?
A different feature	An optional collectible needs its own complete instructions and tests. Agree extra teaching time first.

Expected answers

1. A background changes how the game looks. A moving obstacle changes the route the player can take, so it changes what the player must do.
2. Forever keeps the journey repeating. Without it the Hazard makes one trip and then stops, so the feature only works once.
3. The consequence belongs to Player because Player is the sprite that must move. A check on Hazard that says go to x y moves the Hazard, not the player.
4. In the bug project the contact rule is on Hazard. It moves Hazard to the Player start; Player stays put. The active glide then continues. Rebuild the contact script on Player using touching Hazard, then remove the faulty Hazard contact script.
5. Compare 2-second and 4-second glides from the same positions. Slower is not always easier: it can occupy a passage longer. Choose using repeated results and the pupil's comfort.
6. Outcome 5 needs a feature that changes gameplay, demonstrated by the pupil. A new backdrop, a costume change or a colour edit is decoration and does not meet it.

A likely misconception

Adding a picture is not adding a mechanic. Ask: what must the player now do that they did not do before? If nothing changes, the feature is decoration.

Access and evidence

Record the pupil action and the support used

Flexible support

Current need	Adaptation
Reading or memory	Use short instructions, the reference page and one action at a time.
Communication	Accept speech, pointing, drawing or exact-word scribing.
Keyboard or pointer access	Use an agreed school access device, or let the pupil direct an adult and record who decided and who acted.
Regulation or attention	Offer quiet work, a pause and the same return step.
Ready to transfer	Use Challenge after the common goal. Ask for a reason with a test result.

Pupil voice with a response

Revisit your Week 2 mission idea. Which feature is achievable today? Agree an adaptation and say why.

Ask “What I said, and what it changed”. Listen and reply: use it now, adapt it together, or defer it with a reason and a time.

Individual evidence

AQA 71638 outcome 5: Observe the pupil creating the feature and explaining what the player must now do. Record a before-and-after test. Decoration alone does not meet the outcome.

Pupil action or check	Date and support used
I created a new sprite and named it myself.	
My feature moves or responds on every run, not only once.	
Contact with my feature has a consequence the player can see.	
The four controls, the wall rule and the win still work after my change.	

Record completed outcomes on the AQA summary sheet. Do not infer creation from supplied code.

If a device is unavailable

Sketch a route on plain paper and use a small paper marker for Hazard. Rehearse its effect and the needed blocks. Complete the code and tests on a device.

Shared checks and teaching notes

Week 7 Classroom reference

The common task is a moving Hazard plus a contact consequence. Use the two scripts in the slides. Pupils choose the look, journey and timing. A collectible is a later alternative that needs its own full instructions.

Setup in the supplied maze	Expected result
Flag. Player at (0, 0). Hazard starts (0, -100), glide 2 seconds.	After 1 second Hazard reaches Player. Player returns to (-180, -120).
Flag. Player at (30, 0). Advance 1 second.	Hazard reaches (0, 0), while Player remains (30, 0).
Flag. Watch 2 seconds, then another 2 seconds.	Hazard reaches (0, 100), then returns to (0, -100). It repeats.
Flag after a win. Test a wall and another win.	Message clears, both sprites reset, and earlier rules still work.

The HTML model advances in named time steps so pupils can predict before each move. Scratch glides continuously. Keyboard arrow steps in the model advance 0.1 second; the time button advances 0.5 second.

For a custom maze, choose two endpoints in clear space. Keep the full Hazard costume clear of the Player start and Finish. Adjust the size or journey if every route is blocked. The model uses a small diamond Hazard.

During I Do, pupils select or paint Hazard. During We Do 2, rehearse the contact script on their project. Use Independent Task for completion and testing. If the prerequisite game is unfinished, agree additional construction time before attempting the final assessment.

AQA summary sheets remain the official evidence. Use the final evidence check to locate individual work for outcomes 1 to 6. A saved teacher model cannot establish pupil authorship.